

ORDER NUMBER	0183922
PO NUMBER	373803
ORDER DATE	01/10/24

MATERIAL TEST REPORT

S O L D	DODSON GLOBAL INC P O BOX 330759 HOUSTON TX 77233	S H I P	DODSON GLOBAL 2750 HOLMES ROAD HOUSTON TX 77051
--	---	--	---

ITEMS

Ln#	Qty	Description	Matl Spec	Ht Cd	Mill Heat Number	CE
020	10	8" 900 RF WN STD MSS A694 F52	A694N-16/MSS-SP-44-19	1E164N	NU4000018182	0.43
021	10	8" 900 RF WN XH MSS A694 F52	A694N-16/MSS-SP-44-19	1E164N	NU4000018182	0.43
023	5	12" 900 RF WN STD MSS A694 F52	A694N-16/MSS-SP-44-19	1F536N	NU4000018815	0.41

CHEMICAL PROPERTIES

* Ladle Chemical Properties Indicated with an Asterisk (*)

Ht Cd	C	Si	Mn	P	S	Cr	Al	Cu	Ni	Mo	V	Nb	Sn	N	B	Pb	Ti
1E164N	0.1750	0.2250	1.1800	0.0095	0.0053	0.0827	0.0303	0.1800	0.0632	0.0272	0.0597	0.0000	0.0000	0.0000	0.0004	0.0000	0.0020
*	0.1800	0.2240	1.2200	0.0070	0.0020	0.0800	0.0250	0.1800	0.0600	0.0200	0.0610	0.0000	0.0040	0.0000	0.0004	0.0000	0.0010
1E164N	0.1750	0.2250	1.1800	0.0095	0.0053	0.0827	0.0303	0.1800	0.0632	0.0272	0.0597	0.0000	0.0000	0.0000	0.0004	0.0000	0.0020
*	0.1800	0.2240	1.2200	0.0070	0.0020	0.0800	0.0250	0.1800	0.0600	0.0200	0.0610	0.0000	0.0040	0.0000	0.0004	0.0000	0.0010
1F536N	0.1520	0.2770	1.1500	0.0098	0.0048	0.0771	0.0234	0.2090	0.0672	0.0250	0.0618	0.0000	0.0000	0.0000	0.0002	0.0000	0.0025
*	0.1500	0.2760	1.2100	0.0080	0.0030	0.1000	0.0180	0.2100	0.0700	0.0200	0.0690	0.0000	0.0060	0.0167	0.0003	0.0000	0.0020

PHYSICAL PROPERTIES

* Tensile and Yield Strengths are shown in psi

Ht Cd	Tensl*	Yield*	%Elong	%R.A.	HBW	Charpy	Ft-Lbs	Lat Expsn	%-Shear	Test Temp
1E164N	78000	53800	36.10	74.19	156/156	TRANSV.	38/50/60	.018"/.026"/.033"	25/30/35	-50 F
1E164N	78000	53800	36.10	74.19	156/156	TRANSV.	38/50/60	.018"/.026"/.033"	25/30/35	-50 F
1F536N	73400	54100	35.40	75.58	149/149	TRANSV.	97/94/68	.049"/.053"/.042"	60/55/55	-50 F

NOTES

Ht Cd	Country of Melt	Note
1E164N	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL
1E164N	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL
1F536N	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL

All flanges bearing the "BOLTEx" name are machined in the United States at our Houston Texas facility. Flanges marked with "MADE IN USA" or "USA" are made from steel melted in the United States. Country of melt is shown in the notes section above. Billet used in the manufacturing process has been fully killed. Boltex flanges are manufactured, sampled, tested and inspected per MSS SP-44 2019 PSL1.

It is guaranteed that all Boltex forged steel flanges marked B16 or MSS, adhere to and comply with all dimensions and tolerances in accordance with specification ASME B16.36-2020, ASME B16.5-2020/ASME B16.47-2020 and MSS SP-44-2019 PSL1 as applicable for flanges. Boltex flanges have a surface finish of 125/250 roughness average (Ra) on all full face and raised face finishes.

We certify our flanges marked B16 or MSS, are capable of passing a hydrostatic test compatible with their rating and all test results and process information contained herein are correct and true as contained in company records. The flanges meet the requirements of the latest edition of NACE MR0175/ISO15156 and MR0103/ISO17945.

Our quality system has been registered by DNV-GL to ISO 9001, Certificate No.: 142240-2013-AQ-USA-ANAB. This MTR complies with EN 10204-3.1. Our flanges satisfy the material requirements for PED 2014/68/EU ANNEX 1, Section 4.3, Type 3.1. Notwithstanding the absence of a signature, the organization submitting either a printed certificate or an EDI transmitted certificate is responsible for report content or the report as per ASTM A961-21 Section 19.5.

NOTICE: Any unauthorized downloading, reproduction, distribution, or use of this electronic media is hereby prohibited. Violators shall be prosecuted to the fullest extent of the law.


Quality Assurance Manager